

# SEC P vs NP — Wall atlas

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 23:39 UTC

## SEC P vs NP — Wall atlas

Compiled 2026-05-18 23:39 UTC. 36 total barrier rejections logged.

This is the **map of known walls** encountered by the barrier filter. Each entry below was a proposed conjecture that, according to dual-LLM agreement, would fall to one of the four classical barriers. Patterns in this data tell us where the proposer keeps tripping.

### Barrier-hit counts

| Barrier        | Rejections |
|----------------|------------|
| ALGEBRIZATION  | 13         |
| NATURAL_PROOFS | 13         |
| RELATIVIZATION | 7          |
| KARP_LIPTON    | 3          |

### Rejections by field\_A (top 10)

| field_A                                                               | total | primary barrier |
|-----------------------------------------------------------------------|-------|-----------------|
| Abstract recursion theory                                             | 4     | RELATIVIZATION  |
| Boolean function analysis                                             | 4     | RELATIVIZATION  |
| Arithmetic interactive proofs                                         | 4     | ALGEBRIZATION   |
| Meta-complexity                                                       | 3     | KARP_LIPTON     |
| {“framework_name”:<br>“Tropical Circuit Weight<br>Analysis (TCWA)”    | 2     | ALGEBRIZATION   |
| Representation Theory of<br>Symmetric Groups                          | 2     | ALGEBRIZATION   |
| Abstract recursion                                                    | 1     | RELATIVIZATION  |
| Schur-Horn majorization<br>theory (doubly-stochastic<br>/ diagonal    | 1     | NATURAL_PROOFS  |
| Quantile statistics of<br>random subcube<br>restrictions (order-st    | 1     | NATURAL_PROOFS  |
| Coarse Geometric<br>Karchmer-Wigderson<br>(CG-KW); coarse<br>geometry | 1     | NATURAL_PROOFS  |

## Recent rejections (last 15)

- [2026-04-29 23:34 UTC] ALGEBRIZATION (conf 0.90) — Tropical Circuit Weight Accumulation Bound
- [2026-04-30 15:04 UTC] ALGEBRIZATION (conf 0.90) — Kronecker Coefficients of Rectangular  $GL_n$ -Representations Lower-Bound Determinant
- [2026-05-01 07:13 UTC] ALGEBRIZATION (conf 0.90) — Tropical Circuit Homotopy Stability under Weight Accumulation
- [2026-05-01 07:17 UTC] ALGEBRIZATION (conf 0.90) — Orbit Closure Dimension vs. Boolean Circuit Size
- [2026-05-01 17:19 UTC] ALGEBRIZATION (conf 0.85) — Moment Matrix Eigenvalue Gap and SOS Degree for Max-CUT
- [2026-05-07 22:35 UTC] NATURAL\_PROOFS (conf 0.78) — Leinster Magnitude of Hamming-Embedded Communication Matrices Lower-Bounds  $R(DIS)$
- [2026-05-08 00:42 UTC] NATURAL\_PROOFS (conf 0.88) — Stern-Brocot Rank of Truth Tables Lifts Average-to-Worst MCSP
- [2026-05-08 06:17 UTC] NATURAL\_PROOFS (conf 0.85) — Plunnecke Doubling of Top-Fourier Spectrum Bounds  $ACC_0$  Size
- [2026-05-08 09:57 UTC] NATURAL\_PROOFS (conf 0.78) — Bivariate Mahler Measure of Sign Polynomial Lower-Bounds  $R(DIS)$
- [2026-05-08 11:12 UTC] NATURAL\_PROOFS (conf 0.78) — Lasserre Moment-Rank Gap of Accept Set Bounds  $AC_0$  PARITY Size
- [2026-05-08 15:19 UTC] ALGEBRIZATION (conf 0.95) — Bounded Arithmetic Complexity of Algebraic Closure in Finite Fields
- [2026-05-10 13:12 UTC] ALGEBRIZATION (conf 0.95) — Schur Coefficient Gap in Permanent vs Determinant Decompositions
- [2026-05-10 16:34 UTC] ALGEBRIZATION (conf 0.95) — Schur Coefficient Exponential Gap in 3-CNF Permanent vs Determinant
- [2026-05-17 03:06 UTC] NATURAL\_PROOFS (conf 0.78) — Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds  $DISJ_{CC\_R}$
- [2026-05-17 10:38 UTC] ALGEBRIZATION (conf 0.78) — Cubic-Form Lie-Stabilizer Codim Lower-Bounds  $DISJ$  Communication